Gagan Kavaraganahalli Prasanna

Data Engineer | Boston, Massachusetts | (413)-466-2586 | kpgagan20@gmail.com | linkedin.com/in/gagan-k-p | github.com

Education

University of Massachusetts Amherst — Amherst, MA
Master of Science in Business Analytics — GPA: 4.0 — *Sep 2024 – Feb 2026*

Visvesvaraya Technological University — Bangalore, India
Bachelor of Engineering in Information Science & Engineering — GPA: 3.3 — *Aug 2017 – Jun 2021*

Technical Skills

Data Ingestion & Pipelines: ETL/ELT Design, Apache Spark (PySpark), Batch Ingestion, REST API Ingestion, Schema Validation, Medallion Architecture, Apache Airflow
Languages: SQL (PostgreSQL, Snowflake, SQL Server), Python (Pandas, NumPy, PySpark), Java
Cloud & Data Warehouse: Snowflake, Databricks, AWS (S3, RDS, EC2, Lambda), Azure Data Lake, Docker
Backend & Systems: Spring Boot, Microservices, RESTful APIs, TDD (JUnit, Mockito), SonarQube, Maven
Analytics & Monitoring: Tableau (LOD Expressions), KPI Design, Git, JIRA, Confluence, Agile/Scrum

Experience

Accenture — Bangalore, India
Data Engineer — *Apr 2022 – Jun 2024*

- **Reduced client data onboarding latency by 70%** by engineering 10+ high-throughput RESTful ingestion services on Spring Boot microservices that unified provisioning data across AWS, Azure, and GCP, with schema contracts validated via Swagger.
- **Eliminated 10+ hours per week of manual data handling** by designing **idempotent ELT pipelines in Snowflake** that ingested and standardized data from 4 disparate sources into analytics-ready datasets, enabling downstream reporting at scale.
- **Achieved 92% pipeline test coverage and reduced production defects by 20%** by applying TDD with JUnit and Mockito, enforcing SonarQube quality gates, and standardizing incoming payloads via Java regex-based **schema validation**.
- **Accelerated environment reproducibility** by containerizing all ingestion service dependencies in Docker, enabling consistent on-demand spin-up and eliminating integration environment drift across 5+ Agile sprints.

DXC Technology — Bangalore, India
Data Engineer — *Aug 2021 – Apr 2022*

- **Accelerated incident detection by 40% and recovered 20 hours per week** by engineering a SQL log ingestion and transformation pipeline that automated recurring error pattern detection across 50+ application servers, replacing manual triage with a structured anomaly workflow.
- **Reduced unplanned operational downtime by 20%** by designing release-level performance benchmark pipelines that flagged regression risks before deployment, and building Tableau LOD dashboards surfacing uptime anomalies from the transformed metrics layer.

DXC Technology — Bangalore, India
Analyst Intern — *Apr 2021 – Jun 2021*

- **Reduced defect reopens by 30% and query duplication by 20%** by analyzing 2,000+ QA log records in SQL to isolate 8 recurring defect patterns and building reusable data access layers across 3 teams.

Projects

WorldCupIQ: FIFA 2026 Predictive Analytics & AI Platform — Link
Python, dlt, DuckDB, MinIO, XGBoost, ChromaDB, Ollama, MCP, Streamlit — *Jun 2026*

- **Architected a zero-cloud-cost medallion lakehouse** (MinIO + DuckDB) with **dlt**-powered API ingestion and web scrapers, transforming **49K+ international matches and 1,249 player records** into leakage-free ML feature tables.
- **Trained an XGBoost Poisson goal model** that beat an Elo baseline on 4,500+ held-out matches, powering a **vectorized Monte Carlo engine** simulating **10,000 full-tournament runs in seconds** for champion odds.
- **Integrated AI via RAG and MCP**: ChromaDB-indexed press coverage + Ollama generate player narratives, and a **MCP server** exposes the warehouse, vector store, and simulator as guarded tools for LLM clients and an in-app NL query chat.

ClinicalPulse: Patient Readmission Risk Intelligence Platform — Link
Python, XGBoost, SHAP, LangChain, ChromaDB, Streamlit — *Apr 2026*

- **Engineered a healthcare data pipeline over 100K patient encounters with 50+ clinical features**, structuring raw records into an ML-ready analytical dataset and training an XGBoost readmission model (AUC = 0.70) with SHAP explainability.
- **Built a RAG clinical search engine** (LangChain + ChromaDB) over 6 medical specialty note corpora, enabling semantic retrieval of readmission risk patterns to augment structured ML predictions.